Ibtida Bin Ahmed

ibtida01.ahmed@gmail.com — +8801521768986 — github.com/ibtida01 — LinkedIn — Ibtida01.github.io

Education

Bangladesh University of Engineering and Technology (BUET) Expected Graduation: Late 2027
B.Sc. in Computer Science and Engineering Dhaka, Bangladesh

Technical Skills

Competitive Programming	Codeforces Max Rating: 1551 (Specialist)
Languages	C, C++, Java, Python, JavaScript, C#, LaTeX
Frameworks	React, Node.js, Express.js, Spring Boot
Databases	PostgreSQL, MongoDB
Libraries	STL, NumPy, Pandas, Matplotlib, PyTorch
Tools	Git, GitHub Actions

Projects

Project DRISHTI
Achievement: Innovation Fair 2026,presented to Mr. Mahdi Amin,Advisor of PM of Bangladesh
Project DRISTI — AI-Powered Diabetic Retinopathy Screening

Developed an EfficientNet-B6-based deep learning pipeline for automated diabetic retinopathy detection and severity grading in rural healthcare settings. Implemented fundus image preprocessing using Circular Fundus Isolation,Ben Graham Gaussian subtraction, and CLAHE for enhanced image quality. Optimized continuous regression predictions using Nelder-Mead thresholding,achieving a Quadratic Weighted Kappa (QWK) score of 0.9102. Enabled rapid, on-site DR screening and triage to support early diagnosis and reduce dependency on ophthalmologist availability.

ONE — Unified Citizen Service Platform
Achievement: FRICTION Hackathon, Noverse Inc. — 2nd Runners-Up

Tech Stack: React 18, Node.js, Express.js, PostgreSQL, Groq (Llama 3.3 70B), GitHub Actions.

Shobarkhamar — AI Fish & Poultry Disease Detection
Live: shobarkhamar.vercel.app

Tech Stack: React, FastAPI, PostgreSQL, EfficientNet-B4, EfficientNetV2, Railway, Vercel.

Breast Cancer Risk Prediction (PRIME+) — Ongoing
Python, PyTorch, NumPy, Pandas, Matplotlib.

MangOH Railway Fault Detection Using IoT — Ongoing
Using Sierra Wireless MangOH Development Board.

Status: Under consideration.

Tech Stack: CLI-based tools, Operating Systems.

LASER Tag Hardware Project
Microcontroller-based hardware project.

Used ESP32 microcontroller to build a laser tag system with infrared communication and scoring mechanism.

Tech Stack: C++, JavaScript, HTML.

Achievements

- **FRICTION Hackathon (Noverse Inc.)** — 2nd Runners-Up (Team kiloByte) Solo run, 2026
- **IEEE CUET Sciblitz AI Hackathon 2026** — 5th place, 2026
- **PSTU National Hackathon 2026** — finalist, 2026
- **IUT Datathon — Bengali LLM Hallucination Detection Challenge** — 11th (Team name: DataDriven), July 2026
- **CUET CSE Fest** — Finalist (Solo), 2025
- **NASA Space Apps Challenge 2025** — Regional Champion, 2025
- **BUET CSE Fest Hackathon** — Rising Team (Team kiloByte), 2023
- **BUET CSE Fest CTF** — Rising Team, 5th Place (Team Hellicon), 2023

Languages

- Bangla (Native)
- English (Professional Proficiency)